Database Systems & Design
Chapter 5: Relational Database Normalization

1. Introduction to Normalization
Normalization is the systematic approach of decomposing database tables to eliminate data redundancy and unc

Key Goals of Normalization:
- Minimizing duplicate data across tables
- Ensuring data integrity and consistency
- Enforcing functional dependencies and referential constraints
- Optimizing database storage and transactional integrity

2. Database Anomalies
Without normalization, databases suffer from three classical anomalies:
- Insertion Anomaly: Inability to record certain information without adding extraneous unrelated data.
- Deletion Anomaly: Loss of critical unintended data when deleting a record.
- Update Anomaly: Inconsistent updates across duplicate rows leading to data divergence.

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3. First Normal Form (1NF)
A relation is in First Normal Form (1NF) if and only if:
- Every attribute contains only atomic (indivisible) values.
- There are no repeating groups or multivalued attributes.
- Each record is uniquely identified by a primary key.

4. Second Normal Form (2NF)
A relation is in Second Normal Form (2NF) if:
- It meets all criteria for First Normal Form (1NF).
- No non-prime attribute is partially dependent on any candidate key.
In other words, all non-key attributes must be fully functionally dependent on the entire primary key, not a proper

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5. Third Normal Form (3NF)
A relation is in Third Normal Form (3NF) if:
- It satisfies Second Normal Form (2NF).
- No non-prime attribute is transitively dependent on the primary key.
Transitive dependency exists when X -> Y and Y -> Z, meaning non-key attribute Z depends on non-key attribute

6. Boyce-Codd Normal Form (BCNF)
Boyce-Codd Normal Form (BCNF) is a stricter version of 3NF.
A relation is in BCNF if and only if for every non-trivial functional dependency X -> Y, X is a superkey of the relati

Comparison Summary:
- 1NF eliminates repeating groups.
- 2NF eliminates partial functional dependencies.
- 3NF eliminates transitive functional dependencies.
- BCNF eliminates remaining functional dependency anomalies where determinant is not a candidate key.